

Get Familiar with Remote Execution and Job Orchestration

It is possible to perform remote execution and job orchestration on a number of virtual machines with software like Fabric and Ansible, as this article explains.



Modern computing environments are highly distributed in nature and comprise many physical and/or virtual machines. In fact, with the advent of cloud computing in the last few years, most of the infrastructure has gone virtual with the running of a large number of VMs (virtual machines) to cater to the needs of applications. The need to execute commands on a number of machines is a very common day-to-day requirement for any modern infrastructure. If the operating system running on the machines is GNU/Linux, which is most often the case these days, then SSH sessions are used to execute the desired remote commands. The SSH command provided by various GNU/Linux distributions, MacOS X and Windows (you need to install the very popular Putty SSH client here) is the most universal way to SSH into remote machines (requires `sshd` running and accepting SSH connections from outside, which is the default in almost all GNU/Linux distributions) in an interactive or batch mode. Using the interactive SSH

sessions is fine when working with a few remote machines, but is not suitable when the need is to remote execute on a number of machines without any human intervention.

The good news is that modern dynamic languages like Python, Ruby and even Perl have some good libraries and frameworks around SSH, which enables remote serial or parallel execution of commands on machine. It provides features like retrying multiple times with some delay in between, skipping unresponsive machines as well. You could also use the Fabric API to create many standalone remote execution tools in Python, taking advantage of the capabilities of both the Python standard library, extra modules and Fabric.

Ansible is the big daddy of Fabric, and it is a very popular choice currently, in the IT world. It is a full-fledged but no-fuss cloud provisioning, configuration management and job orchestration framework created in Python. Ansible further